

CSCI3280: INTRODUCTION TO MULTIMEDIA

Final Project: Build Your AI Travel Agent ✈️

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VECTOR ILLUSTRATION
CYBER ENGINE

The Vision: Redefining Travel Planning

Beyond ChatGPT

Not just a text wrapper, but an active, intelligent agent.

Multimodal Interaction

Talk to it directly, and let it speak back to you naturally.

Actionable

It actively searches, routes, and creates actionable plans for you.



Traditional Planning

Scattered tabs, overwhelming information, and stressful manual scheduling across multiple apps.



Your AI Agent

A unified, clean voice interface that generates structured, personalized itineraries instantly.

Minimum Viable Product (MVP) – The Pass Line



Requirement 1

Modality

Must support Speech & Text input and output seamlessly.

- ✓ Voice Input Recognition
- ✓ Text-to-Speech Output
- ✓ UI for Text Fallback



Requirement 2

Live Search

Must connect to search APIs to fetch real-time data.

- ✓ No pure LLM hallucination
- ✓ Live location data integration
- ✓ Dynamic API fetching



Requirement 3

Core Functions

The agent must handle the entire lifecycle of a trip.

- ✓ **Plan:** Destination itinerary
- ✓ **Introduce:** Attraction details
- ✓ **Route:** Transport planning

How to get an A+? 🌟

Visual Input/Output

Enhance the experience by displaying relevant, high-quality photos of the attractions. Or plan the trip according to input images of attractions.

Advance Route Planning

How to go through all the attractions using the shortest distance

Weather-Aware Planning

Implement dynamic routing that adapts the itinerary based on real-time and forecasted weather conditions.

Other Useful Functions/Features

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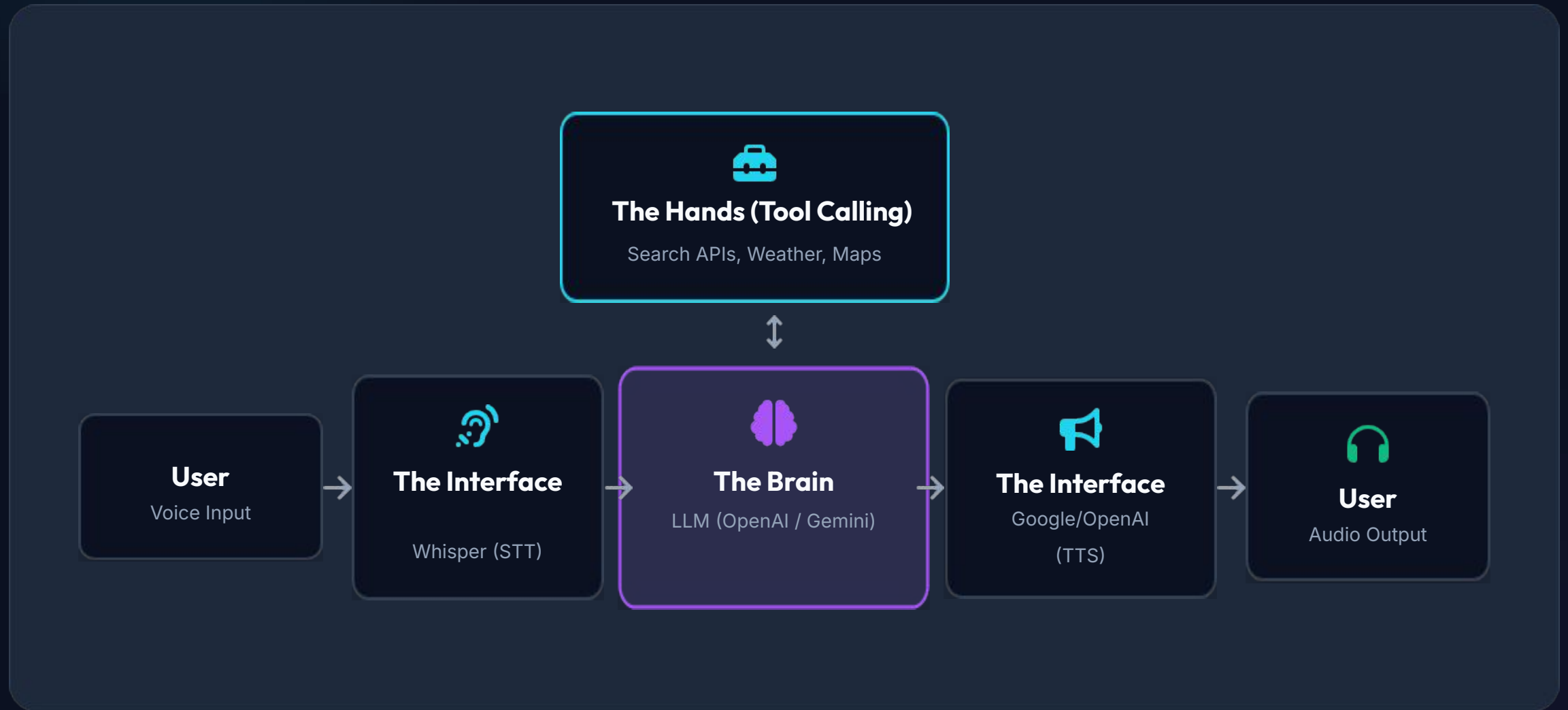
User Profiling (Memory)

Remember user preferences across sessions (e.g., "I hate crowded places", "I love historical museums").

A Commercial Example

imean.ai

How does it work? (Architecture)



We've Got You Covered!



Upcoming Tutorial 1

Learn the core techniques of Prompting for Agents and structuring reliable JSON outputs from LLMs.



Upcoming Tutorial 2

Master Tool Calling fundamentals and learn how to integrate STT/TTS seamlessly into your pipeline.



Resources Provided

We will provide starter code, API wrappers, and token quotas to jumpstart your development process.



Let's build something awesome!

Find and Register Your Team!

Teams are limited to a maximum of 4 members. [Register Link](#) (By 28 Feb).

Grading Details

The grading is divided into three categories:

1. **Basic Functionality (60%):**
 - ASR, LLM, and TTS modules working together.
2. **Extended Functionality (20%):**
 - Useful features, novel design, better user interface, low latency.
3. **Presentation (20%):**
 - Clear, fluent presentation with a demonstration of the system.

Report Requirements

- A **final report** may contain:
 - A description of the system architecture.
 - Explanation of how all the modules are integrated.
 - Detailed technical approach and challenges faced.
 - The introduction and adaptation of the system to your target domain.
 - Code comments and explanations.
 - Specific contributions of each team member in the project.
 - The length of the main report (excluding possible appendix) is at most **4 pages**, academic conference style.

Presentation

- Present your system with a clear, structured explanation.
- Demonstrate the running travel planning agent during the presentation.
- Prepare a presentation for about **8 mins** per group.
- The presentation will held in the lectures of the last week.

Code Submission

- Submit all **source code** of the project.
- Ensure the code is well-organized, commented, and executable.